

Linear ODE

$$y' + p(x)y = Q(x)$$

$$I = e^{\int p(x) dx}$$

$$\frac{d}{dx} [y \cdot e^{\int p(x) dx}] = Q(x) \cdot e^{\int p(x) dx}$$

" Bernoulli eq.

$$y' + p(x)y = Q(x) \cdot y^n \quad (n \neq 0, 1)$$

$$\rightarrow y^n \left[y^{-n} \cdot y' + p(x) \cdot y^{1-n} \right] = Q(x) \quad \text{--- (1)}$$

$$\text{令 } u = y^{1-n}, \quad \frac{du}{dx} = (1-n) \cdot y^{-n} \cdot \frac{dy}{dx}$$

$$\xrightarrow{\text{代回(1)}} \frac{1}{(1-n)} \cdot \frac{du}{dx} + p(x) \cdot u = Q(x)$$

$$\xrightarrow{\times (1-n)} \frac{du}{dx} + \frac{(1-n) \cdot p(x)}{p(x)} \cdot u = \frac{(1-n) \cdot Q(x)}{Q(x)}$$

Wronskian $\rightarrow$ 判斷是 L.I./L.D.

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \quad W(x) \neq 0 \rightarrow \text{L.I.}$$

= 齊次 常係數

Non-Homogeneous eq.

step 1 找 Homogeneous 解

step 2 找 y_p

1. $R(x)$	y_p	2. $y_p = Ae^{rx}$
K	A	$y_p' = rAe^{rx}$
e^{ax}	Ae^{ax}	$y_p'' = r^2 A e^{ax}$
$\cos ax$	$A \cos ax + B \sin ax$	代回原式
$\sin ax$		
$a_n x^n + \dots + a_1 x + a_0$	$A_n x^n + A_{n-1} x^{n-1} + \dots + A_0$	

Δ 若 y_p 之假設項與 n 個 L.I.

y_h 重複, y_p 要修正

$$\rightarrow y_p = A x^k e^{mx}$$

$$[\text{Case 1}] m = r \rightarrow y_h = c_1 e^{rx} + c_2 e^{-rx}$$

$$[\text{Case 2}] m = \pm i \rightarrow y_h = c_1 e^{ix} + c_2 e^{-ix}$$

$$[\text{Case 3}] m = 1 \pm 2i \rightarrow y_h = e^{ix} (c_1 \cos 2x + c_2 \sin 2x)$$

<sol 2> 令 $y_p = \phi_1 y_1 + \phi_2 y_2$

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} \phi_1' \\ \phi_2' \end{bmatrix} = \begin{bmatrix} 0 \\ R(x) \end{bmatrix}$$

$$\phi_1' = \frac{\begin{vmatrix} 0 & y_2' \\ R(x) & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$\phi_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & R(x) \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$\phi_1 = \int \phi_1' dx$$

$$\phi_2 = \int \phi_2' dx$$

$$\therefore y_p = \phi_1 y_1 + \phi_2 y_2$$

$$\therefore y = y_h + y_p$$

Cauchy 等變換性

$$\text{令 } t = \ln x \rightarrow x = e^t$$

$$D \equiv \frac{d}{dt}$$

$$x \frac{dy}{dx} = \frac{dy}{dt} = Dy$$

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y = \frac{d^2 y}{dt^2} - \frac{dy}{dt}$$

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y = \frac{d^3 y}{dt^3} - 3 \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt}$$

$$x^4 \frac{d^4 y}{dx^4} = D(D-1)(D-2)(D-3)y =$$

$\rightarrow$ 把它們代回原式 $\Rightarrow$ 回到常係數

最後換回 x

Fourier series

The Fourier series of f on $[-\lambda, \lambda]$ is

$$a_0 + \sum_{n=1}^{\infty} (a_n \cos(\frac{n\pi x}{L}) + b_n \sin(\frac{n\pi x}{L}))$$

$$a_0 = \frac{1}{2\lambda} \int_{-\lambda}^{\lambda} f(t) dt$$

$$a_n = \frac{1}{\lambda} \int_{-\lambda}^{\lambda} f(t) \cdot \cos(\frac{n\pi t}{L}) dx$$

$$b_n = \frac{1}{\lambda} \int_{-\lambda}^{\lambda} f(t) \cdot \sin(\frac{n\pi t}{L}) dx$$

$$[-L, L] = [-2, 2]$$

$$\cos(n\pi x) \rightarrow \text{偶} \quad \sin(n\pi x) = 0 \quad \text{奇} \times \text{偶} = \text{奇}$$

$$\sin(n\pi x) \rightarrow \text{奇} \quad \cos(n\pi x) = (-1)^n$$

$$\text{奇} \times \text{奇} = \text{偶} \quad \text{偶} \times \text{偶} = \text{偶}$$

$$\int_0^2 t \sin(\frac{n\pi t}{2}) dt$$

$$D \quad I$$

$$+ \quad t \quad \sin(\frac{n\pi t}{2})$$

$$- \quad 1 \quad -\frac{2}{n\pi} \cos(\frac{n\pi t}{2})$$

$$+ \quad 0 \quad -\frac{4}{n^2\pi} \sin(\frac{n\pi t}{2})$$

$$= -\left[\frac{2t}{n\pi} \cos(\frac{n\pi t}{2}) + \frac{4}{n^2\pi} \sin(\frac{n\pi t}{2}) \right]_0^2$$

$$\int e^{ax} \cos(bx) dx =$$

$$= \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx))$$

$$\cos(2n\pi) = 1, \sin(2n\pi) = 0$$

$$\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$$

$$(e^{ax})' = ae^{ax}$$

$$(a^x)' = a^x \ln a$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \tan x dx = -\ln|\cos x| + C$$

$$\int \cot x dx = \ln|\sin x| + C$$

$$\int \sec x dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x dx = \ln|\csc x - \cot x| + C$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

(b) $g(y), y' = f(x)$
 $\int g(y) y' dx = \int f(x) dx + C$
 $\rightarrow \int g(y) dy = \int f(x) dx + C$

1) 可分离变量 $\rightarrow$ ① 可直接分离
 $\rightarrow x$ 放一边, y 放一边
 $y' = f(x, y) \rightarrow M(x, y) dx + N(x, y) dy = 0$
 $\rightarrow \frac{dy}{dx} = f(x, y) \rightarrow \int \frac{M(x, y)}{N(x, y)} dx = - \int \frac{N_2(y)}{M_2(y)} dy + C$
 $\rightarrow \int \frac{1}{f(y)} dy = \int f(x) dx + C$

② (1) - 阶齐次 ODE
(2) 型 $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$
(3) 型 $y' = f(ax + by + c)$

③ (1) - 阶齐次 ODE
可设 $y' = f(\frac{y}{x})$ or $y' = f(\frac{y}{x})$, 整理成该型
变量变换 $u = \frac{y}{x} \rightarrow y = ux$
 $\rightarrow dy = xdu + udx$
代回原式
2. 分离变量 (回到 ①)
② (B)
 $t = ax + by + c$
 $dt = adx + bdy$
 $dy = \frac{dt - adx}{b}$
代回原式

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\sinh x)' = \cosh x$$

$$(\cosh x)' = \sinh x$$

$$(\ln x)' = \frac{1}{x}$$

$$(\log_a x)' = \frac{\log_a e}{x}$$

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\cos^{-1} x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2}$$

$$(\cot^{-1} x)' = -\frac{1}{1+x^2}$$

$$\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \sinh^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \cosh^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \sin^2 x dx = \frac{1}{2} x - \frac{1}{4} \sin 2x + C$$

$$\int \cos^2 x dx = \frac{1}{2} x + \frac{1}{4} \sin 2x + C$$

$$\int \tan^2 x dx = \tan x - x + C$$

③ - (2)
 $(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$
 $L_1 = L_2 = 0$ (回到 ③ - (1))
 $C_1 \neq 0$ or $C_2 \neq 0$ $\begin{cases} \Delta \cdot \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \text{有交点} \\ \Delta \cdot \frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow \text{平行} \end{cases}$

Δ
 $a_1x + b_1y + c_1 = 0 \rightarrow a_1x + b_1b + c_1 = 0$ 求交点 $(x, y) = (\alpha, \beta)$
 $a_2x + b_2y + c_2 = 0 \rightarrow a_2\alpha + b_2\beta + c_2 = 0$
令 $x = u + \alpha, dx = du$
 $y = v + \beta, dy = dv$
variable $\uparrow$ 常数
代回原式
 $[a_1(u+\alpha) + b_1(v+\beta) + c_1]du + [a_2(u+\alpha) + b_2(v+\beta) + c_2]dv = 0$
 $\Rightarrow (a_1u + b_1v)du + (a_2u + b_2v)dv = 0$
 $\Rightarrow (a_1 + b_1 \frac{v}{u})du + (a_2 + b_2 \frac{v}{u})dv = 0$
令 $t = \frac{v}{u}$ (回到 变量变换)

$$\int \ln|x| dx \parallel u = \ln x \quad dv = dx$$

$$= x \ln x - x + C$$

Exact $\phi(x, y) = C$
 $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow$ ③ ② $M = \frac{\partial \phi}{\partial x} \quad N = \frac{\partial \phi}{\partial y}$
 $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 0$ or $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = \frac{1}{xy}$ or $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = \frac{1}{xy}$
 $I = \int f(x) dx$
 $I(M(x, y)dy + N(x, y)dx) = 0 \cdot I$
 $M' = \frac{\partial}{\partial x} \left(\frac{1}{xy} \right) \rightarrow \phi(x, y) = x^2 y + \ln|y|$
 $N' = \frac{\partial}{\partial y} \left(\frac{1}{xy} \right) \rightarrow \phi(x, y) = x^2 y + \ln|y|$

51. $f(x, y) = \int M dx + g(y)$
52. $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M dx + g'(y)$
53. $g'(y) = N - \frac{\partial}{\partial y} \int M dx$
54. $\int g'(y) dy = g(y)$

Δ 令 $\begin{cases} a_2x + b_2y = z \rightarrow a_2dx + b_2dy = dz \\ a_1x + b_1y = mz \end{cases} \rightarrow dy = \frac{dz - a_2dx}{b_2}$
代回
 $(mz + c_1)dx + (z + c_2) \left(\frac{dz - a_2dx}{b_2} \right) = 0$
 $\rightarrow dx + \dots dz = 0$
 $\rightarrow \int$